

```
In [1]: #Fawaz Alshahrani  
#2240005571
```

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In [ ]: import cv2  
import numpy as np  
import matplotlib.pyplot as plt  
  
image = cv2.imread('images/Parrot.png')  
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)  
smoothed = cv2.blur(image, (7, 7))  
  
fig, ax = plt.subplots(1, 2, figsize=(10, 5))  
ax[0].imshow(image)  
ax[0].set_title("Original Image")  
ax[0].axis("off")  
  
ax[1].imshow(smoothed)  
ax[1].set_title("Smoothed Image (7x7 Box Filter)")  
ax[1].axis("off")  
  
plt.show()
```

Original Image



Smoothed Image (7x7 Box Filter)



```
In [ ]: import cv2  
import numpy as np  
import matplotlib.pyplot as plt  
  
image = cv2.imread('images/Parrot.png')  
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)  
gaussian_5x5 = cv2.GaussianBlur(image, (5, 5), 0)  
  
gaussian_21x21 = cv2.GaussianBlur(image, (21, 21), 0)  
  
fig, ax = plt.subplots(1, 3, figsize=(15, 5))  
ax[0].imshow(image)  
ax[0].set_title("Original Image")  
ax[0].axis("off")  
  
ax[1].imshow(gaussian_5x5)  
ax[1].set_title("Gaussian Filter (5x5)")  
ax[1].axis("off")  
  
ax[2].imshow(gaussian_21x21)  
ax[2].set_title("Gaussian Filter (21x21)")  
ax[2].axis("off")
```

```
plt.show()
```

Original Image



Gaussian Filter (5x5)



Gaussian Filter (21x21)



```
In [ ]: import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('images/Parrot.png')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
image_float = image.astype(np.float32) / 255.0

laplacian = cv2.Laplacian(image_float, cv2.CV_32F, ksize=3)

sharpened = np.clip(image_float - laplacian, 0, 1)

fig, ax = plt.subplots(1, 3, figsize=(15, 5))
ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(np.clip(laplacian, 0, 1))
ax[1].set_title("Laplacian Filter Output")
ax[1].axis("off")

ax[2].imshow(sharpened)
ax[2].set_title("Sharpened Image (Laplacian)")
ax[2].axis("off")

plt.show()
```

Original Image



Laplacian Filter Output



Sharpened Image (Laplacian)

